Matthew Ju

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Education

University of Illinois Urbana-Champaign **Expected May 2028**
Bachelor of Science in Computer Science & Statistics | GPA: 3.77/4.00 *Champaign, Illinois*
Relevant Coursework: Data Structures & Algorithms, Systems Programming, Database Systems, Software Design

Technical Experience

Corteva **May 2026 – August 2026**
Software Engineering Intern *Johnston, Iowa*

- Refactored **ASP.NET Core** backend code across 13 controllers into a layered service architecture that separates business logic from data access while adding pagination and response DTOs to REST endpoints
- Added **Microsoft Entra ID** JWT authentication to previously unprotected production API endpoints and fixed a file-path traversal vulnerability
- Wrote 8 **Playwright** end-to-end test suites and more than 10 **xUnit** service tests and configured Azure DevOps YAML pipelines to run automated tests as predeployment gates
- Cut **Azure DevOps** CI build time in half through caching and test sharding across Playwright and xUnit test suites
- Integrated external scouting and work-order services into application workflows, handling cancellation and edit events to update local status data

AbbVie **June 2024 – August 2024**
Data Engineering Intern *North Chicago, Illinois*

- Built a **Python** and **Django** data pipeline for industrial sensor experiments, modeling individual runs and metadata as structured records and reducing experimental processing time by 50%
- Implemented CSV persistence and file-handling tools to store and retrieve experimental runs through a command line interface, alongside signal processing and anomaly detection using moving-average and median filters

Projects

PearlMetric | C# / ASP.NET Core, Python / FastAPI, Angular, PostgreSQL, OpenCV, Docker | **Live**

- Built a dental shade tracking MVP with an **ASP.NET Core** gateway, a **FastAPI** image-analysis service using OpenCV for tooth-region segmentation and color comparison, and an Angular frontend
- Implemented 17 authenticated REST endpoints using **EF Core** and a **PostgreSQL** schema with constraints and indexes while keeping image-analysis HTTP calls outside database transactions
- Wrote more than 50 unit and integration tests across C#, Python, and Angular; validated shared JSON contracts with cross-language fixtures and configured **GitHub Actions** to build and test pull requests
- Configured **Docker Compose** for application services, PostgreSQL, and local image storage with Caddy providing TLS, gateway routing, and restrictions on internal routes

ListingLab | FastAPI, Angular, PostgreSQL, Redis, Vercel | **Live**

- Built and deployed an eBay price-research application with **FastAPI** and Angular that uses concurrent API requests to retrieve and deduplicate up to 400 listings per search
- Implemented **Redis** result caching, per-IP rate limiting, and a lock to coordinate OAuth token refreshes across serverless instances along with exponential backoff and jitter for upstream errors
- Implemented IQR outlier filtering and log-gap price clustering with interactive ECharts visualizations linked to listing-table rows and URL-based search state for shareable searches

Skills

Languages: Python, C#, TypeScript, C++, Java, SQL, R
Frameworks & Libraries: ASP.NET Core, EF Core, FastAPI, Django, Angular, React, Next.js, pandas, NumPy
Databases & Caching: PostgreSQL, Azure SQL, Redis
Tools & Platforms: Git, Docker, Azure DevOps, GitHub Actions, Vercel
Testing: Playwright, xUnit, pytest